

Revenue Forecasting for Fashion E-Commerce

Datathon 2026 Round 1 | R2U Team | Technical Report

Team R2U — HCMC University of Technology

Vương Thành Đạt (Data Engineering) | Trần Nguyễn Hưng (Visualisation) | Phạm Phương Thảo (Modelling)

Abstract

This report presents the R2U team's solution for the Datathon 2026 Round 1 revenue-forecasting challenge on a fashion e-commerce dataset. The workflow covers: (i) data engineering — normalising and merging 14 CSV sources into a single feature table; (ii) four-layer analytics — descriptive, diagnostic, predictive, and prescriptive insights via 12+ charts; and (iii) gradient-boosting ensemble (LightGBM, XGBoost, CatBoost) with seed averaging, optimised for daily Revenue prediction on 2023–2024 test data. The design emphasises strict leakage prevention, reproducibility (fixed seeds), and explainability via SHAP.

Part 2: Data Engineering & Analytics

2.1 Pipeline Overview

The raw dataset comprises **14 CSV files** in four groups: *Master Data* (products, customers, promotions, geography), *Transaction Data* (orders, order_items, payments, shipments, returns, reviews), *Operational Data* (inventory, web_traffic), and *Analytical Data* (sales, sample_submission). The pre-processing pipeline operates in five stages:

- ID normalisation:** All ID columns and zip cast to `str`; missing promo codes filled with "NONE".
- Time enrichment:** `order_date` expanded into year, month, quarter, dayofweek, and `is_weekend` flags.
- Missing-value handling:** Numeric negatives converted to NaN then filled with column median; categorical values filled with "Unknown".
- Pre-merge aggregation:** Payments grouped by `order_id` (total paid, payment count, primary method); inventory keyed on (`product_id`, `YearMonth`).
- Master Merge** (10 LEFT JOINS, anchored on `order_items`): orders → products → customers → geography → shipments → returns → reviews → inventory → web_traffic → business logic (shipping fee allocation, traffic imputation).

Leakage prevention: All joins use only information available at order-placement time; future-dated fields (`delivery_date`, `return_date`, `review_date`) are carried as

post-hoc labels, never used as training features in Part 3.

2.2 Feature Engineering

From the merged table, the following derived columns are computed:

- Financial:**

$\text{Gross_Rev} = \text{qty} \times \text{unit_price},$

$\text{Net_Rev} = \text{Gross_Rev} - \text{discount},$

$\text{Total_Cost} = \text{qty} \times \text{COGS} + \text{ship_fee},$

$\text{Target_Profit} = \text{Net_Rev} - \text{Total_Cost} - \text{Refund}_{[\text{is_returned}=1]}.$

- Logistics:**

$\text{Lead_Time} = \text{delivery_date} - \text{order_date} \quad (\text{days}),$

$\text{Shipping_Efficiency} = \text{Lead_Time} / \text{ship_fee}.$

- Customer:**

`Customer_Tenure`: days since first purchase,

`Order_Frequency`: cumulative order count per customer.

2.3 Main Findings

Descriptive. Revenue is concentrated: the top ~20% of products account for 80% of net revenue (Pareto analysis). Electronics and Apparel lead in gross revenue, but their profit margins differ significantly — high-revenue categories may carry thin margins.

Diagnostic. A waterfall decomposition reveals the revenue erosion chain: Gross Revenue → (–) Discount → (–) Refunds → (+) Shipping → Net Revenue. Discount is the single largest leakage point. The promo-vs-no-promo boxplot shows that average order value (AOV) is *not* significantly higher for promoted orders, suggesting promotions may be subsidising customers who would have purchased anyway.

Predictive. Cohort retention matrices show customer attrition is steepest in months 2–3 after first purchase. ARPU varies substantially across regions, identifying high-value geographic targets.

Prescriptive. See Table 1.

Table 1: Prescriptive summary — key findings and actions.

Area	Finding	Recommendation
Products	Top 20% SKUs drive 80% revenue; several high-revenue SKUs have sub-10% margin	Focus upsell / ad spend on core SKUs; re-price or discontinue low-margin items
Promotions	Discount is the largest revenue drain; AOV not lifted by promos	Shift from blanket discounts to targeted BOGO/free-ship on high-CLV segments
Inventory	Products cluster in stockout-risk and overstock quadrants simultaneously	Adopt demand-signal reorder triggers; liquidate overstock via flash sales
Quality	High-volume SKUs with poor ratings and elevated return rates act as hidden loss centres	Audit supplier quality for flagged SKUs; enforce review-response SLAs

Part 2 Notebook — Source & Charts | https://github.com/vngthdat206/R2U-Datathon2026-TheGridbreakers/blob/main/Notebooks/R2U_Part2.ipynb

Part 3: Methodology, Models & Results

3.1 Problem Statement

Predict **daily Revenue** (and COGS) for a fashion e-commerce platform on the 2023–2024 test horizon, using only: `sales.csv` (historical Revenue/COGS), `orders.csv` (order counts), and `web_traffic.csv` (sessions, page views, bounce rate). No external data or target leakage is permitted.

3.2 Feature Engineering

- Calendar & seasonality:** month, day, dayofweek, dayofyear, weekofyear, quarter; binary flags: `is_weekend`, `is_month_start/end`, `is_quarter_start/end`.
- Event signals:** `is_double_day` (11/11, 12/12, ...); `is_payday` (1st, 5th, 15th, 25th of month); `days_to_next` / `days_since_prev` double-day (clipped at 30).
- External signals (leading indicators):** daily order count aggregated from `orders.csv`; `sessions`, `unique_visitors`, `page_views`, `bounce_rate` from `web_traffic.csv`.
- Historical proxy features (no leakage):** Median of Revenue, `orders_count`, `sessions`, `page_views`, `bounce_rate` grouped by (month, dayofweek), (day), and (weekofyear), computed exclusively on 2021–2022

data. These act as learned baselines (“*model memory*”).
Total feature set: **30 features**.

3.3 Training Strategy

Time-based split: Train $\leq$ 2021-12-31; Validation = full year 2022; Test $\geq$ 2023-01-01. This replicates real deployment conditions — the model never sees future data.
Seed averaging: Each model is trained on 5 seeds {42, 2026, 8888, 777, 999} and predictions are averaged, reducing variance without cross-validation overhead.

Ensemble formula:

$$\hat{y} = 0.40 \hat{y}_{\text{LGB}} + 0.20 \hat{y}_{\text{XGB}} + 0.40 \hat{y}_{\text{CB}}$$

Weights were chosen empirically: LightGBM and CatBoost showed lower validation MAE and received higher weight.

Table 2: Hyperparameters (identical across 5 seeds).

Model	Iter	LR	Depth	Leaves	Sub.
LightGBM	800	0.012	7	45	0.80
XGBoost	800	0.012	6	—	0.80
CatBoost	850	0.015	6	—	0.80

3.4 Validation Performance (Year 2022)

Table 3: Metric comparison on held-out 2022 validation set.

Model	MAE	RMSE	R^2
LightGBM (avg)	computed — see notebook output		
XGBoost (avg)	computed — see notebook output		
CatBoost (avg)	computed — see notebook output		
Ensemble	best; printed as <code>pred_val_final</code> metrics		

3.5 Explainability — Feature Importance & SHAP

LightGBM feature importance (averaged over 5 seeds) and SHAP `mean(|value|)` consistently identify four driver groups:

- Proxy Revenue features** (`proxy_Revenue_m_dow`, `proxy_Revenue_day`, `proxy_Revenue_week`) — top SHAP contributors; anchor predictions to historical baseline.
- Calendar / seasonality** (month, dayofweek, weekofyear) — capture strong weekly and monthly revenue cycles.
- Event markers** (`is_double_day`, `is_payday`, `days_to_next_double`) — model learns spike behaviour around mega-sale dates and paydays.

4. **Traffic proxies** (`proxy_sessions_*`, `proxy_page_views_*`) — confirm sessions as a leading indicator of daily revenue.

Business interpretation. Revenue is highly predictable under normal conditions (stable historical patterns). Shocks to traffic quality (elevated bounce rate) or supply (stockout events) are the primary sources of forecast error. Recommended actions: (i) run flash sales precisely on double-day and payday events; (ii) optimise landing pages to reduce bounce rate; (iii) build weekly/monthly campaign calendars aligned with proven recurring patterns.

Part 3 Notebook — Models & Submission | https://github.com/vngthdat206/R2U-Datathon2026-TheGridbreakers/blob/main/Notebooks/R2U_Part3.ipynb

Conclusion

The R2U pipeline transforms 14 heterogeneous CSV files into a unified analytical backbone, enabling both granular business diagnosis and accurate revenue forecasting.

Key insights: (i) revenue is structurally concentrated in a small SKU core — targeted investment outperforms broad discounting; (ii) customer retention falls sharply in months 2–3, making early re-engagement campaigns high-ROI; (iii) stockout and overstock risk co-exist in inventory, addressable via demand-signal reordering; (iv) daily revenue is forecastable with the ensemble model by exploiting calendar patterns, event spikes, and traffic signals — with no external data needed.

Future work: incorporate real-time inventory feeds as model features; extend SHAP analysis to customer-segment level; explore online learning to adapt to distribution shift in the test period.

GitHub Repository (full source, notebooks, submission)

<https://github.com/vngthdat206/R2U-Datathon2026-TheGridbreakers>